Run report: ctrlK-q3-split-run-20261024T2152Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (exhaustive, q = 3) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: ctrlK-q3-split 20261024T2152Z
git HEAD: 0fb870d0ee76ba6a1b059e9e34c869518fb412b0
tool: examples/genus4_n4_census.rs sha256 f4c7f7355c838b6f112e09cb412fd3353d087309962c64cd39ad49d79f4c9dad
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 a95b1e35748cefe23a044d8cfd9e059b723cf71fdbc4af56db71257da42c8b33
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/ctrlK-q3-split-run-20261024T2152Z CENSUS 3 split exh 1 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 68
```

tool_log.txt

```
split q=3: samples=43046721 (exhaustive) res_shaped=179184 allowed_trace_candidates=23040 smooth_prime_order_hits=2304 smooth_fraction_among_prime_cands=0.1000
smooth_fraction_other_res_shaped(sub-sample 2000)=412/2000 secs=67.9
split q=3: total prime-order mass estimate 1.000 +- 0.021 (hit rate per sampled form 5.352e-5)
```

summary.txt

```
split q=3: samples=43046721 (exhaustive) res_shaped=179184 allowed_trace_candidates=23040 smooth_prime_order_hits=2304 smooth_fraction_among_prime_cands=0.1000
smooth_fraction_other_res_shaped(sub-sample 2000)=412/2000 secs=67.9
split q=3: total prime-order mass estimate 1.000 +- 0.021 (hit rate per sampled form 5.352e-5)
```

cand_t_hist.csv (first 12 of 16 rows)

type	t	candidates
split	−15	2304
split	−11	4608
split	−9	2304
split	−8	12672
split	−6	9216
split	−5	2304
split	−3	11520
split	−2	25344
split	−1	20736
split	0	28848
split	1	28800
split	2	4608

hits.csv (first 12 of 23040 rows)

type	code	t	nE	smooth
split	769526	−1	83	false
split	770270	−1	83	false
split	773192	−1	83	false
split	774323	11	71	true

split	774751	−1	83	false
split	776345	−1	83	false
split	776931	−1	83	false
split	777355	11	71	true
split	777706	−1	83	false
split	779931	−1	83	false
split	780110	11	71	true
split	780696	−1	83	false

mass_by_t.csv (first 1 of 1 rows)

type	t	nE	smooth_hits	mass_estimate	mass_stderr
split	11	71	2304	1.0000	0.0208

Graph

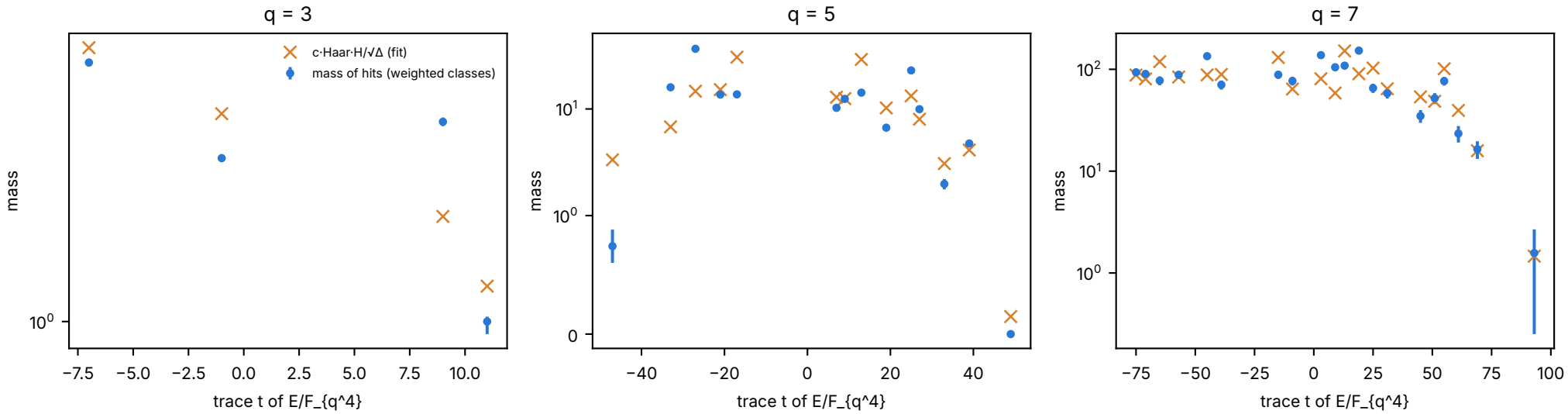


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.